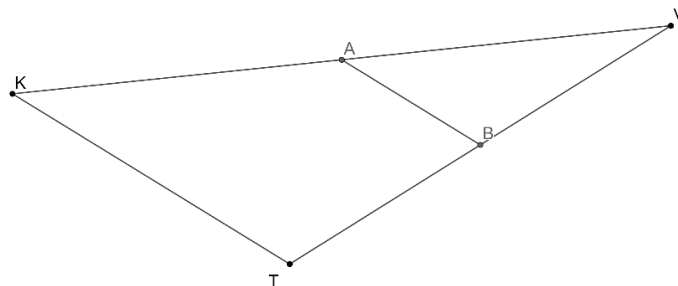


Geometry
Unit 11
Lesson 8 Practice

Name **Answer Key**
Date _____
Period _____

Use this information to answer questions 1 through 5.

Triangle KVT is shown, where $\overline{AB}$ is the midsegment. $KT = 2x + 4$, $AB = \frac{1}{3}x + 4$, $AV = \frac{3}{2}y + 3.5$, $KA = \frac{1}{2}y + 7.5$, $VT = 5z + 3$, and $TB = \frac{3}{4}x + 6$.



1. Find the value of x .

$$\begin{aligned} \frac{3}{2} \left[2 \left(\frac{1}{3}x + 4 \right) \right] &= 2x + 4 \\ x + 12 &= 3x + 6 \\ 2x &= 6 \\ x &= 3 \end{aligned}$$

2. Find the value of y .

$$\begin{aligned} 2 \left[\frac{1}{2}y + 7.5 \right] &= \frac{3}{2}y + 3.5 \\ y + 15 &= 3y + 7 \\ 2y &= 8 \\ y &= 4 \end{aligned}$$

3. Find the value of z .

$$\begin{aligned} 2 \left(\frac{3}{4}x + 6 \right) &= 5z + 3 \\ \frac{3}{2}(3) + 12 &= 5z + 3 \\ 5z &= \frac{27}{2} \\ z &= 2.7 \end{aligned}$$

4. Find the length of the midsegment, $\overline{AB}$.

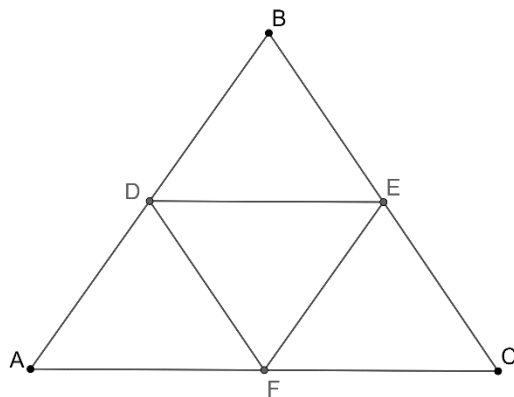
$$AB = \frac{1}{3}(3) + 4 = 5$$

5. Find the length of $\overline{KV}$.

$$\begin{aligned} KV &= KA + AV \\ KV &= \left(\frac{1}{2}(4) + 7.5 \right) + \left(\frac{3}{2}(4) + 3.5 \right) = 19 \end{aligned}$$

Use this information to answer questions 6 through 10.

Triangle ABC is shown, where D, E , and F are midpoints. $DE = 2x - 4$, $AC = \frac{1}{2}x + 13$, $DF = \frac{3}{2}x$, $BC = 5x - 12$, $FE = 3x - 11$ and $AB = \frac{5}{3}x + 4$.



6. Find the value of x . $2(2x - 4) = \frac{1}{2}x + 13$

$$4x - 8 = \frac{1}{2}x + 13$$

$$8x - 16 = x + 26$$

$$7x = 42$$

$$x = 6$$

7. Find the length of $\overline{AB}$.

$$AB = \frac{5}{3}(6) + 4 = 14$$

8. Find the length of $\overline{BC}$.

$$BC = 5(6) - 12 = 18$$

9. Find the length of $\overline{AC}$.

$$AC = \frac{1}{2}(6) + 13 = 16$$

10. Find the perimeter of triangle DEF .

$$P = 14 + 18 + 16 = 48$$